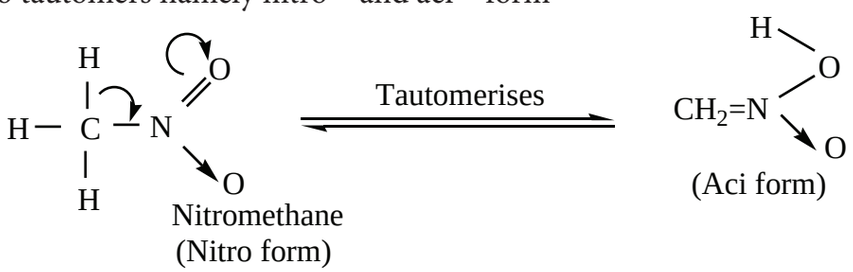
13.1.3 ISOMERISM

Nitroalkanes exhibit chain and position isomerism among their own class and functional isomerism with alkyl nitrites and special type tautomerism can also exist in nitro alkanes having an α-H atom. For example, nitro compounds having the molecular formula $C_4H_9NO_2$ exhibit the following isomerisms.

Isomerism	Structural formula of isomers
Chain isomerism: They differ in the length of carbon chain.	$CH_3CH_2CH_2CH_2 - NO_2$ and $CH_3CHCH_2 - NO_2$ (with CH_3 branch) 1 - nitrobutane 2 - methyl - 1-nitropropane
Position isomerism: They differ in the position of nitro group.	$CH_3CH_2CH_2CH_2 - NO_2$ and $CH_3CH(NO_2)CH_2CH_3$ 1 - nitrobutane 2 - nitrobutane
Functional isomerism: Nitroalkanes exhibit functional isomerism with alkylnitrites	$CH_3CH_2CH_2CH_2 - NO_2$ and $CH_3CH_2CH_2CH_2 - O - N = O$ 1 - nitrobutane butyl nitrite

Tautomerism: Primary and secondary nitroalkanes, having α-H , also show an equilibrium mixture of two tautomers namely nitro – and aci – form



Nitromethane (Nitro form) (Aci form)

Tertiary nitro alkanes donot exhibit tautomerism due to absence of α-H atom.

S.No.	Nitro form	Aci – form
1.	Less acidic	More acidic
2.	Dissolves in NaOH slowly	Dissolves in NaOH instantly
3.	Decolourises $FeCl_3$ solution	With $FeCl_3$ gives reddish brown colour
4.	Electrical conductivity is low	Electrical conductivity is high

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